1 $\overline{A \circ B} = \bar{A} \bullet \hat{B}$

$$\overline{A \circ B} = \overline{(A \ominus B) \oplus B} \tag{1a}$$
$$= \overline{A \ominus B} \ominus \hat{B} \tag{1b}$$
$$= (\bar{A} \oplus \hat{B}) \ominus \hat{B} \tag{1c}$$
$$= \bar{A} \bullet \hat{B} \tag{1d}$$

2 $\overline{A \oplus B} = \bar{A} \ominus \hat{B}$

By definition:

$$A \oplus B = \{\mathbf{x} | (\hat{B})_{\mathbf{x}} \cap A \neq \phi\} \tag{2}$$
$$A \ominus B = \{\mathbf{w} | B_{\mathbf{w}} \subseteq A\} \tag{3}$$

Taking the complement of the dilation:

$$\overline{A \oplus B} = \overline{\{\mathbf{x} | (\hat{B})_{\mathbf{x}} \cap A \neq \phi\}} \tag{4a}$$
$$= \{\mathbf{x} | (\hat{B})_{\mathbf{x}} \cap A = \phi\} \tag{4b}$$

If $(\hat{B})_{\mathbf{x}} \cap A = \phi$, then $(\hat{B})_{\mathbf{x}} \subseteq \bar{A}$
Therefore:

$$\overline{A \oplus B} = \{\mathbf{x} | (\hat{B})_{\mathbf{x}} \cap A = \phi\} \tag{4c}$$
$$= \{\mathbf{x} | (\hat{B})_{\mathbf{x}} \subseteq \bar{A}\} \tag{4d}$$
$$= \bar{A} \ominus \hat{B} \tag{4e}$$